

Advanced Radio License Notes

Oskar Meszaros (z5636164)

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1 Introduction

This are the notes from following the advanced course on HamRadio.au.

2 Electronics and Circuit Theory

2.1 Impedance

2.1.1 Introduction

Impedance is how a circuit stores or resists AC (or any changing) energy. If the impedances within a circuit aren't matched, then power gets reflected back instead of going where it is needed.

Impedance is the combination of resistance and reactance. Resistance is the restriction of the "flow" of energy, and converts that restricted energy to heat. This is measured in Ohms Ω . Reactance can be thought of as a flywheel, it stores energy temporarily and releases it. This is what capacitors and inductors do. This is also measured in Ohms. This means that impedance is the total opposition to current.

Why this matters. The transmitter expects to see 50Ω of impedance. The coax is also 50Ω . If the antenna isn't also close to 50Ω , then the power from the transmitter gets reflected, which is what SWR (standard wave ratio) measures.

2.1.2 Reactance

There are two types of reactance. One is found in inductors, and the other in capacitors.

Inductors (any coil of wire) opposes changes in current, i.e. their reactance increases with frequency

$$X_L = 2\pi fL.$$

This means that the higher the frequency, the more and inductor blocks the signal. This is why they are used in low-pass filters, they all for low frequencies but block high ones.

Capacitors opposes changes in voltage. Their reactance decreases with frequency

$$X_C = \frac{1}{2\pi fC}.$$

The higher the frequency, the easier it passes through the capacitors. This is why a capacitor in series blocks DC but passes RF.

2.1.3 Impedance formula

We can then write impedance as

$$Z = R + jX,$$

where R is the resistance, X is the impedance, and j is used in place of i for imaginary numbers due to i being used for current. The resistance part is where the energy is consumed, and the reactive part is where the energy is stored and returned. The total impedance is then given by the magnitude of this complex number

$$|Z| = \sqrt{R^2 + X^2}$$

Example. An antenna has $R = 50\Omega$ and $X = 25\Omega$. The total impedance magnitude is

$$\begin{aligned} Z &= \sqrt{2500 + 625} \\ &= \sqrt{3125} \\ &\approx 56\Omega \end{aligned}$$

2.2 Resonance

2.2.1 Introduction

When an inductor and capacitor are combined, at one specific frequency their reactances become equal and therefore cancel each other out. This is called resonance, and when this happens the circuit behaves as purely resistive. This is how a single frequency is selected while rejecting everything else.

Radio Example. Tuned circuits within a receiver use this resonance to select a station to hear, and reject everything else. Antennas are resonant at a specific frequency, which is where it works best.

The formula to find the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC}}.$$

This says that for a given inductor L and capacitor C , there's exactly one frequency where they resonate. This is an important formula.

2.2.2 Series and Parallel Resonance

Series and parallel arrangements of L and C result in opposite behaviour at resonance.

Table 1: Behaviour of circuits at resonance		
	Series LC	Parallel LC
Impedance	Minimum (just R)	Maximum
Current	Maximum	Minimum from source
Used for	Passing a desired frequency (band-pass)	Blocking a desired frequency (or selecting it) (band-stop)

Radio Example. A series-resonant circuit in an antenna trap passed the trap's frequency through, while a parallel-resonant circuit in a receiver's IF (intermediate frequency) stage has high-impedance at the desired frequency, which develops a large signal voltage across it.

2.2.3 Q-Factor

The Q-factor describes how selective a resonant circuit is. A higher Q means a very sharp tuning (with narrow bandwidth), and low Q means a very broad tuning (wide bandwidth)

$$Q = \frac{f_r}{\text{Bandwidth}}$$

For a series RLC , $Q = \frac{X_L}{R}$, so less resistance means higher Q.

Bandwidth Rule. Bandwidth is $\frac{f_r}{Q}$, so a circuit with $Q = 100$ at 7 MHz has a bandwidth of 70 kHz, which is enough to cover a portion of the 40m band. A crystal filter with $Q = 50000$ at 9 MHz has a bandwidth of only 180 Hz, which is perfect for CW.

2.3 Filters

2.3.1 Introduction

Filters are extremely important in radio. They are what selects the frequencies that get let through and what get blocked. Without filters, a receiver would hear literally everything at the same time, and the transmitter would spray harmonics all over the spectrum (which happens to be illegal).

2.3.2 Types of Filter

There are four basic types of filter.

Table 2: Basic types of filter

Filter Type	Behaviour	Where it is in radios
Low-pass	Passes low frequencies, blocks high frequencies	After the transmitter PA (power amplifier). It blocks harmonics before reaching the antenna
High-pass	Passes high frequencies, block low frequencies	At a TV antenna input. It blocks HF signals, but lets VHF/UHF TV through
Band-pass	Passes a specific range, blocks everything else	IF (intermediate frequency) filters in a receiver. It selects the signal bandwidth that you want.
Band-stop (notch)	Blocks a specific range, passes everything else	Notch filter to remove an interfering carrier.

2.3.3 LC Filter

Filters use inductors and capacitors in combinations to make a specific filter. From above, we know

- Inductors block high frequencies (reactance increases with frequency)
- Capacitors block low frequencies (reactance decreases with frequency)

Low-pass Filter. A low-pass filter wants to block high frequencies, and allow low frequencies through. This means that the circuit would put an inductor in series (to block high frequencies), and capacitors to ground (short high frequencies to ground).

2.3.4 Filter Steepness

In practice, filters can't instantly go from passing a frequency to blocking another. The change is gradual, and this is affected by the number of LC sections (called a pole). Each pole adds 20 dB/decade of rolloff. The higher this number, the more steep the dropoff, as it is saying that per decade (1 decade is a ratio of 10 between two numbers on a logarithmic scale), the scale drops by 20 dB.

Example. Consider a transmitter that produces a 7 MHz signal. The 2nd harmonic at 14 MHz and 3rd harmonic at 21 MHz must be suppressed. A good 5-element low-pass filter with a cutoff just about 7 MHz can suppress the 2nd harmonic by 40+ dB and the 3rd by 60+ dB. This is the difference between a clean signal and an interference complaint.

A typical transmitter low-pass filter has 5-poles (100 dB/decade, a steep slope). For critical applications, a 7-pole filter is used (140 dB/decade, a very steep slope).

2.3.5 Crystal Filters

Crystal filters have very high Q factors (10 000 to 100 000+). Multiple crystals combined in a so-called ladder filter give the sharp selectivity for IF stages

- 2.4 kHz bandwidth - for SSB reception
- 500 Hz bandwidth - for CW reception
- 6 kHz bandwidth - for AM reception

In modern radios however, crystals are being replaced with DSP (digital signal processing) filters, as they allow changing the bandwidth on the fly.

2.3.6 Filter Response Shapes

Different mathematical designs give different trade-offs. Butterworth designs have a smooth, flat passband with no ripple. This is used where consistent gain across the passband matters. Chebyshev has a steeper rolloff than Butterworth, but some ripple in the passband. This is used where sharp selectivity matters more than a perfectly flat response.

2.4 Amplifier Classes

2.4.1 Introduction

The final stage of a transmitter is a power amplifier. This amplifier will have some bias, and depending on how that bias is configured changes two important things. First is the linearity - how faithfully does the amplifier reproduce the signal shape (critical for SSB and AM). The second is efficiency - how much of the DC power becomes RF (all excess is heat).

2.4.2 Classes

These are the different classes of amplifiers

Important. Class C amplifiers are not suitable for SSB or AM. Both of these carry information in the variation in amplitude, so when class C clips the signal information is destroyed. It is fine for FM and CW as these modes have a constant amplitude.

Table 3: Amplifier classes

Class	How it works	Efficiency	Linearity	Used for
A	Transistor conducts all the time (360°). Always on	25 – 50%	Excellent	Low-power stages, driver stages
AB	Conducts most of the time (180°-360°). Slightly biased below full-on	50 – 65%	Very good	Standard choice for SSB power amplifiers
B	Conducts exactly half the time (180°). Needs push-pull pairs	65 – 78%	OK	Push-pull audio amps
C	Conducts less than half the time (<180°). Very "punchy"	75 – 85%	Poor	FM and CW only

Most HF transceivers use Class AB for the final PA. It is linear enough to handle SSB without excessive distortion, and efficient enough that a 400W output doesn't require a large heatsink. If a Class AB amp is overdriven (i.e. mic gain is too high), it starts producing intermodulation distortion. This is a "splatter" that interferes with adjacent frequencies.

2.4.3 Decibels

Decibels allows for working with ratios by simply adding and subtracting instead of multiplying and dividing.

Table 4: Shortcuts for working with decibels

dB	Power ratio	Voltage ratio	Notes
+3 dB	$\times 2$	$\times 1.41$	Double the power
+6 dB	$\times 4$	$\times 2$	Double the voltage
+10 dB	$\times 10$	$\times 3.16$	Ten times the power
+20 dB	$\times 100$	$\times 10$	Ten times the voltage
-3 dB	$\times 0.5$	$\times 0.71$	Half the power

Example. An amplifier has 13 dB of gain. That is $10 + 3 = \times 10 \times \times 2 = \times 20$ power. Put in 20W, get out 400W.

Info. The factor is 10 for power, 20 for voltage. That is because power is proportional to the voltage squared.

Power formula: $G_{dB} = 10 \log_{10}(P_{out}/P_{in})$.

Voltage formula: $G_{dB} = 20 \log_{10}(P_{out}/P_{in})$.

2.5 Oscillators

2.5.1 Introduction

An oscillator generates a continuous AC signal at a specific frequency. This is what the transmitter uses to generate the carrier signal, and the receiver needs one to convert signals to the IF. An oscillator is an amplifier where some of the output is fed back into the input in-phase (positive feedback).

In order for this to work correctly, two criteria must be met (known as the Barkhausen criteria). The first is that the loop gain must be greater than or equal to 1. The fed-back signal must be at least as strong as what is needed at the input. The second criteria is that the fed-back signal must be in phase so that it can reinforce the original signal.

2.5.2 Common Types

Colpitts Oscillator: uses two capacitors and one inductor. The two capacitors form a voltage divider for the feedback. It is very popular in VFOs (Variable Frequency Oscillators) as it is stable and easy to tune by varying one of the capacitors.

$$f = \frac{1}{2\pi\sqrt{L \cdot \frac{C_1 C_2}{C_1 + C_2}}}$$

Hartley Oscillator: uses one capacitor and a tapped inductor (or two inductors). The tapped inductor provides the feedback. This is less common in modern equipment.

Crystal Oscillator: uses a quartz crystal as the frequency determining element. The crystal has an extremely high Q (10 000+), making it incredibly stable. It is used for reference oscillators in synthesisers, BFO (Beat Frequency Oscillator) in CW/SSB receivers, and any application where frequency accuracy is critical.

Radio Connection. When tuning a HF radio, usually you are not changing the oscillator directly, rather modern radios use a PLL (Phase-Locked Loop) synthesiser. This is a crystal oscillator that provides the stable reference, and the PLL multiplies it to the frequency that you want.

2.5.3 PLL Frequency Synthesiser

The PLL is how modern radios generate precise, tunable frequencies. In simple terms:

1. A crystal reference provides an accurate, stable signal (e.g. 10 kHz)
2. A VCO (Voltage-Controlled Oscillator) generates the output frequency
3. A divider divides the VCO output down
4. A phase detector compares the divided output with the reference
5. If they don't match, it adjusts the VCO until they do.

$$f_{\text{out}} = N \times f_{\text{ref}}$$

Example. Consider a reference $f_{\text{ref}} = 10$ kHz and $N = 1450$. The output is then $f_{\text{out}} = 14.500$ MHz (20 metres). Changing N to 1451 gives $f_{\text{ref}} = 14.510$ MHz. This is how a tuning knob works in 10 kHz steps.

2.5.4 Mixers

A mixer takes two signals and produces their sum and difference frequencies

$$f_{\text{out}} = f_1 + f_2 \quad \text{and} \quad f_{\text{out}} = f_1 - f_2.$$

This is how receivers convert the signal you want to the IF, and how transmitters shift the generated signal to the operating frequency. The best type of mixer is the double-balanced mixer. This makes filtering much easier as it suppresses both input signals at the output, passing mainly the sum and difference products.

2.6 AGC and Power Supplies

2.6.1 Introduction

AGC (automatic gain control) automatically adjusts the gain of a receiver so that the audio level remains roughly constant. It works by

1. The receiver measures the strength of the incoming signal

2. This creates a control voltage
3. The control voltage adjusts the of the RF and IF amplifier stages

This allows for less gain with strong signals, and more gain with weaker signals. AGC can be either fast or slow. Fast AGC has a quick attack and release, which is best for CW and digital modes, and allows for quick recovery between elements. Slow AGC still has quick attack, however the release is slowed. This is best for SSB voice, and doesn't "pump" between syllables. Pumping is volume surging between words.

2.6.2 Power Supplies for Radio

Most amateur HF transceivers need a regulated 13.8V DC supply. There are two main types of power supply.

Linear power supply: simple and electrically quiet. Works by dropping excess voltage across a regulator, which turns into heat. The efficiency is dependent on the input and output voltages, i.e. $\text{efficiency} = V_{\text{out}}/V_{\text{in}}$. The result of all this is that it is preferred for HF because of low noise, however requires a large mains transformer.

Switch-mode power supply (SMPS): must more efficient and lighter, but can generate RF interference. It works by rapidly switching DC on and off (50-500 kHz), which is then usually passed through a filter to reduce noise. This is significantly cheaper and lighter than linear power supplied.

EMC Consideration. If a receiver has buzzing or whining that changes with tuning, the SMPS might be the cause. Ferrite chokes on the DC power leads dampen this, and keep the power leads away from the antenna feedline.

3 Transmitters and Receivers

3.1 The Superheterodyne Receiver

Trying to make a very narrow filter that can be tuned across the entire HF spectrum (3-30 MHz) would result in having to retune the filter each time the frequency is change. Superheterodyne (superhet) receivers solve this problem by instead of adjusting the filter, it modifies the signal such that it falls within the filter.

A superhet receiver is broken down into stages

1. RF Preselector: a broad filter that roughly selects the band of interest, and rejects signals that are obviously out of band.
2. Mixer + Local Oscillator (LO): The mixer combines the signal with the LO. This transforms the signal to match the IF. For example, if you are tuned to 14.200 MHz and the IF is 9 MHz, then the LO runs at 23.200 MHz ($23.200 - 14.200 = 9.000$ MHz)
3. IF Filter: this is fixed at 9 MHz (or whatever IF), and doesn't change when you tune. It provides the sharp selectivity required. Crystal or DSP filters are used here to determine the bandwidth.
4. IF Amplifier: This boosts the filtered signal (controlled by AGC).
5. Detector: Converts the IF signal back to audio. For SSB, this is a product detector (mixer with a BFO).
6. Audio Amplifier: Drives the output audio device.

3.1.1 The Image Problem

Because the mixer doesn't know which signal produced the 9 MHz difference, there is a second "image" of the signal placed at 2 times the IF. For example, if the LO is at 23.2 MHz then both a signal at 14.2 MHz and a signal at 32.2 MHz produce a 9 MHz signal.

$$f_{\text{image}} = f_{\text{signal}} \pm 2 \times f_{\text{IF}}$$

The RF preselector must be good enough to reject the image. A higher IF makes this easier as the separation between the frequencies is higher. High-end receivers combat this by first converting to a high IF (i.e. 45-70 MHz) to create the separation between the signal and the image, then converts to the final IF (9 MHz or 455 MHz), which allows for narrow and affordable crystal filters.

3.2 SSB

3.2.1 Introduction

Regular AM transmits a carrier plus two sidebands, but since the carrier carries no information and the sidebands carry the same information this is wasteful. Instead, SSB (single sideband modulation) removes the carrier and one sideband, which saves power and bandwidth. Generally, the process is

1. Audio amplifier: the amplified output of the microphone
2. Balanced modulator: mixes the audio with a carrier frequency. It still generates both sidebands, but suppresses the carrier. This is now called DSB-SC (Double Sideband, Suppressed Carrier)
3. Crystal filter: a very sharp band-pass filter that passes one of the sidebands and rejects the other
4. Mixer: shift the SSB signal from the filter up to the operating frequency
5. Power amplifier (Class AB): boosts the signal to the desired output power
6. Low-pass filter: removes the harmonics before transmitting

Convention. The global convention on which sideband is used is dependant on the frequency. When it is less than 10 MHz, the LSB (lower sideband) is used, and for 10 MHz and above, USB (upper sideband) is used.

3.2.2 The Phasing Method

Instead of using an expensive crystal oscillator, this method uses phase shifting to cancel one sideband mathematically. It requires precise 90° shifts in both the audio and RF paths. This is popular in SDR (software defined radio) as phase shifting is easy to do in software.

3.2.3 Balanced Modulator

The balanced modulator is a mixer that is specifically designed so that the carrier leaks through as little as possible. The output contains the two sidebands, but almost no carrier. That is why it is called balanced.

3.3 FM

3.3.1 Introduction

Whilst SSB (and AM) modulates the amplitude of the signal to carry voice, FM varies the frequency, and the amplitude remains constant. FM is used for most VHF/UHF communication (2m and 70cm repeaters), as it is simpler, more noise resistant, and sounds better than SSB for local communications.

For FM, there are three important values. The first is deviation (δf), and is a measure of how far the frequency swings from the carrier. For amateur narrow-band FM, this is ± 5 kHz. The next value is modulation index β , and this is the deviation divided by the audio frequency $\beta = \delta f / f_m$. The final value is the bandwidth (Carson's rule), and is given by

$$\text{BW} \approx 2 (\delta f + f_m).$$

Example. With a ± 5 kHz deviation and 3 kHz maximum audio, bandwidth is given by $2(5 + 3) = 16$ kHz. This is why FM channels on 2m are spaced 12.5 or 25 kHz apart.

One of the biggest advantages of FM is that when two stations transmit on the same frequency, the stronger signal wins completely. The weaker signal is suppressed, not mixed in with the strong one.

Info. The capture effect is why FM works so well for repeaters. If two station accidentally transmit at the same time, you hear the stronger one clearly and not a mix of both. With SSB, you'd hear both simultaneously.

3.3.2 Pre- and De-emphasis

FM noise gets worse at higher audio frequencies. To prevent this, transmitters (pre-emphasis) boost high audio frequencies before transmission, and receivers (de-emphasis) cuts high audio frequencies after demodulation. The boost and cut cancel out for the unwanted signal, but the noise (which was not boosted) gets cut. This results in better signal-to-noise.

3.4 Receiver Performance

3.4.1 Introduction

On a busy HF band, a receiver might be dealing with signals ranging from barely detectable (-130 dBm), to extremely strong (-10 dBm), which is a range of over 100 million to one. Good receivers are able to handle both simultaneously.

3.4.2 Sensitivity

Sensitivity is limited by noise. Every receiver generates some internal noise, and the signal must be stronger than this noise floor to be heard. The noise figure of the front is the key specification, as it tells you how much noise the receiver adds compared to the perfect receiver.

Friis Formula. The noise figure of the first stage dominates, because noise from later stages gets divided by the gain of earlier stages. This is why a low-noise preamp at the antenna makes a big difference.

$$NF_{\text{total}} \approx NF_1 + \frac{NF_2 - 1}{G_1} + \frac{NF_3 - 1}{G_1 G_2} + \dots$$

3.4.3 Intermodal Distortion (IMD)

When two strong signals hit a receiver, any non-linearity creates phantom signals at new frequencies. The most troublesome are third-order products

$$f_{\text{IMD}} = 2f_1 - f_2 \quad \text{and} \quad 2f_2 - f_1.$$

This falls close to the original signals and cannot be filtered out.

Example. Two strong stations on 14.200 and 14.210 MHz create a phantom signal at $2(14.200) - 14.210 = 14.190$ MHz. If you are trying to hear a weaker station at the frequency, you would only be able to hear the IMD product.

IP3 (third-order intercept point) is the key specification from strong signal handling. Higher IP3 is better. This is the theoretical point where the IMD products would equal the wanted signal. This is never reached in practice.

3.4.4 Reciprocal Mixing

In the real world, no oscillator is perfect, they all have some sort of phase noise (random frequency jitter). When a strong off-channel signal mixes with this phase noise, it creates at the IF that buries weak signals. This means that a better oscillator gives better reception, and why more expensive radios perform better on crowded bands.

3.4.5 Dynamic Range

Dynamic range is the difference between the weakest signal the receiver can detect and the strongest signal it can handle without distortion. More dynamic range means better performance on busy bands. A receiver with 90 dB of dynamic range can simultaneously handle signals differing by 90 dB, which is about 1 billion to one in power.

4 Propagation

4.1 The Ionosphere

4.1.1 Introduction

The ionosphere is a layer of charged gas in the upper atmosphere (60-500 km), which is caused by ultraviolet radiation from Sol stripping away electrons. Because of this charge, HF signals are able to bounce off of this layer and travel thousands of kilometres.

4.1.2 The Layers

There are three main layers in the ionosphere.

The D Layer (60-90km) absorbs signals, but only exists during the daylight. It mostly affects signals below 10 MHz, but reflects barely any signals.

The E Layer (90-150km) is able to reflect signals up to 10 MHz (roughly) during the day, but also disappears during the night.

The F Layer (150-500km) is the most important layer for HF communication. During the day it is separated into a F1 and F2 layer, however recombines during the night to a single F layer. The F2 layer is the highest and has the most ionisation, and hence capable of the longest skip distances.

The day/night cycle of HF. During the day, the D layer absorbs low-band signals (80m, 40m), but the F2 layer supports higher bands (20m, 15m, 10m). At night, the D layer vanishes opening up the low bands, while the F layer weakens, which closes the high bands. This affects what bands to use at what times.

4.2 Selecting the Right Frequency

4.2.1 Introduction

For any given path between two stations, there's a range of frequencies that will work. Too high of a frequency and the signal will punch through the ionosphere, too low and the D layer absorbs it.

4.2.2 Maximum Usable Frequency (MUF)

The MUF is the highest frequency the ionosphere will return to Earth for a specific path. Above this, signal passes straight through into space. This frequency varies with time of day, season, solar cycle, and path length.

4.2.3 Minimum Usable Frequency (LUF)

This is the lowest frequency that provides a usable signal. Below this, the D layer absorption is too strong. The LUF is higher during the day and lower during the night.

4.2.4 Frequency of Optimum Traffic (FOT)

This is the most optimum frequency, given by

$$\text{FOT} = 0.85 \times \text{MUF}$$

This is the frequency for the most reliable communication.

Table 5: The best bands for different times		
Time	Best Bands	Notes
Daytime	20m, 17m, 15m, 12m, 10m	High MUF, D-Layer absorbs low bands
Evening	40m, 30m, 20m	D Layer fading, F layer still good
Night	160m, 80m, 40m	No D layer absorption, low MUF
Sunrise/sunset	160m, 80m (grey line)	Special propagation, see below.

4.2.5 Skip Distance and Skip Zone

When a signal goes up to the ionosphere and comes back down, it obviously does not come back down directly next to the transmitter. We call the minimum distance where the skywave signal returns to Earth the skip distance, and the area between where the ground wave fades out and where the skywave returns the skip zone (also known as dead zone). In general, a higher frequency corresponds to a longer skip distance. On 10m, the skip might be 1000+ km, where as 40m it might be 300km.

4.3 Special Propagation

4.3.1 Sporadic E

Sometimes a random patches of ionisation in the E layer can reflect signals well into the VHF range, especially on 6 metres (50 MHz), and even 2m. This is most common in the summer, but is very unpredictable. Single hop distances can be 500-2300 km.

4.3.2 Grey-Line Propagation

The grey-line is the sunset/sunrise boundary on Earth. Along this line, the D layer is either yet to form (sunrise), or has just disappeared (sunset), but the F layer still has enough ionisation to reflect signals. This creates a low-absorption "duct" that reflects signals on the low bands. This is a brief window, and both the transmitting and receiving stations must be on or near the grey line.

4.3.3 Tropospheric Ducting

Not all propagation has to involve the ionosphere. Temperature inversions in the lower atmosphere (troposphere, 0-12km) can create ducts that trap VHF/UHF signals. This is most common over water or flat terrain, and is often accompanied by fog or haze conditions. It can extend VHF/UHF range to 1000+ km.

4.3.4 Earth-Moon-Earth (EME/Moonbounce)

It is possible to bounce signal off of Lunar. This requires high-gain antennas, high power, and very sensitive receivers. The path loss is very high (approx 250+ dB). This is common on 144 MHz, 432 MHz, and 1296 MHz.

4.3.5 Satellite Communication

Amateur radio satellites (OSCAR series, etc) provide a relay capability. This can be difficult due to Doppler shift, and the need to track the satellite and continuously adjust frequency.